

tuple ()

- only 2 func index(value) count(value)- tuple not changable - duplicates allow - indexing & slicing allow but we can't change it

```
In [1]: t=()  
t
```

```
Out[1]: ()
```

```
In [2]: t1=(92,4,5,3)  
t1
```

```
Out[2]: (92, 4, 5, 3)
```

```
In [3]: t2=(4,5.5,False,'Pragati',(4+7j))  
t2
```

```
Out[3]: (4, 5.5, False, 'Pragati', (4+7j))
```

```
In [5]: print(t)  
print(t1)  
print(t2)
```

```
()  
(92, 4, 5, 3)  
(4, 5.5, False, 'Pragati', (4+7j))
```

```
In [6]: t1
```

```
Out[6]: (92, 4, 5, 3)
```

```
In [7]: t1.count(5)
```

```
Out[7]: 1
```

```
In [8]: t1.index(3)
```

```
Out[8]: 3
```

```
In [48]: t1[3]      # indexing allow
```

```
Out[48]: 3
```

```
In [49]: t1[3]=10    # we can't change it
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[49], line 1  
----> 1 t1[3]=10  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [51]: t1
```

```
Out[51]: (92, 4, 5, 3)
```

```
In [50]: t1[1:4]
```

```
Out[50]: (4, 5, 3)
```

```
In [9]: t1.append(10)
        t1                                # append not allow
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[9], line 1
----> 1 t1.append(10)
      2 t1

AttributeError: 'tuple' object has no attribute 'append'
```

```
In [13]: t1[2]=133                        # tuple not changable
        t1
```

```
-----
TypeError                                    Traceback (most recent call last)
Cell In[13], line 1
----> 1 t1[2]=133
      2 t1

TypeError: 'tuple' object does not support item assignment
```

```
In [16]: t4=t1                            # duplicate allow
        t4
```

```
Out[16]: (92, 4, 5, 3)
```

```
In [17]: t1
```

```
Out[17]: (92, 4, 5, 3)
```

```
In [18]: t1.pop()                        # pop() not allow
```

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[18], line 1
----> 1 t1.pop()

AttributeError: 'tuple' object has no attribute 'pop'
```

```
In [19]: t5=t1*2                          # repeate all elements
        t5                                # duplicate allow
```

```
Out[19]: (92, 4, 5, 3, 92, 4, 5, 3)
```

set {}

```
In [ ]: - pop()
        - add()
        - discard()
```

- empty {} called dict - if similar datatype, it is in ordered different datatype, it is unordered - duplicates not allowed - only 1 argument is allowed in add() func - index & slicing not allowed bcoz elem. not in fix ordered, they are unordered - pop() gives random value given in the set pop(index) --> error - only 1 argument allowed - discard() --> never error discard(value) --> remove that value - remove() --> error

```
In [20]: s={}
         s
```

```
Out[20]: {}
```

```
In [23]: s1={8,10,2,4,5,7}           # same datatype in order
         s1
```

```
Out[23]: {2, 4, 5, 7, 8, 10}
```

```
In [22]: s2={3,4,6,4.4,(4+5j),True}   # diff datatype,unordered
         s2
```

```
Out[22]: {(4+5j), 3, 4, 4.4, 6, True}
```

```
In [24]: print(s)
         print(s1)
         print(s2)
```

```
{ }
{2, 4, 5, 7, 8, 10}
{True, 3, 4, 4.4, 6, (4+5j)}
```

```
In [25]: s1
```

```
Out[25]: {2, 4, 5, 7, 8, 10}
```

```
In [27]: s1.add(1)
         s1
```

```
Out[27]: {1, 2, 4, 5, 7, 8, 10}
```

```
In [28]: s1
```

```
Out[28]: {1, 2, 4, 5, 7, 8, 10}
```

```
In [29]: s1.add(3,2)           only 1 argument allow
         s1
```

```
-----
TypeError                                 Traceback (most recent call last)
Cell In[29], line 1
----> 1 s1.add(3,2)
      2 s1

TypeError: set.add() takes exactly one argument (2 given)
```

```
In [30]: s1
```

```
Out[30]: {1, 2, 4, 5, 7, 8, 10}
```

```
In [32]: s1.clear()
         s1           # all value remove in set
```

```
Out[32]: set()
```

```
In [34]: s2
```

```
Out[34]: {(4+5j), 3, 4, 4.4, 6, True}
```

```
In [35]: s2.copy()
```

```
Out[35]: {(4+5j), 3, 4, 4.4, 6, True}
```

```
In [39]: s2.discard(3)    # remove 3 from set
s2
```

```
Out[39]: {(4+5j), 4, 4.4, 6, True}
```

```
In [40]: s2[3]    # index not allow
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[40], line 1
----> 1 s2[3]

TypeError: 'set' object is not subscriptable
```

```
In [41]: s2[2:4]    # slicing not allow bcoz it's unordered
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[41], line 1
----> 1 s2[2:4]

TypeError: 'set' object is not subscriptable
```

```
In [43]: s1={1,3,5,30,24,56}
s1
```

```
Out[43]: {1, 3, 5, 24, 30, 56}
```

```
In [44]: s1.pop()    # gives any random no.
```

```
Out[44]: 1
```

```
In [45]: s1.pop(3)    # index not allow bcoz it's unordered
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[45], line 1
----> 1 s1.pop(3)

TypeError: set.pop() takes no arguments (1 given)
```

```
In [46]: s1
```

```
Out[46]: {3, 5, 24, 30, 56}
```

```
In [47]: s1.remove(4)
```

```
-----  
KeyError                                Traceback (most recent call last)  
Cell In[47], line 1  
----> 1 s1.remove(4)  
  
KeyError: 4
```

In []: